

Web Technology



Javascript – Object

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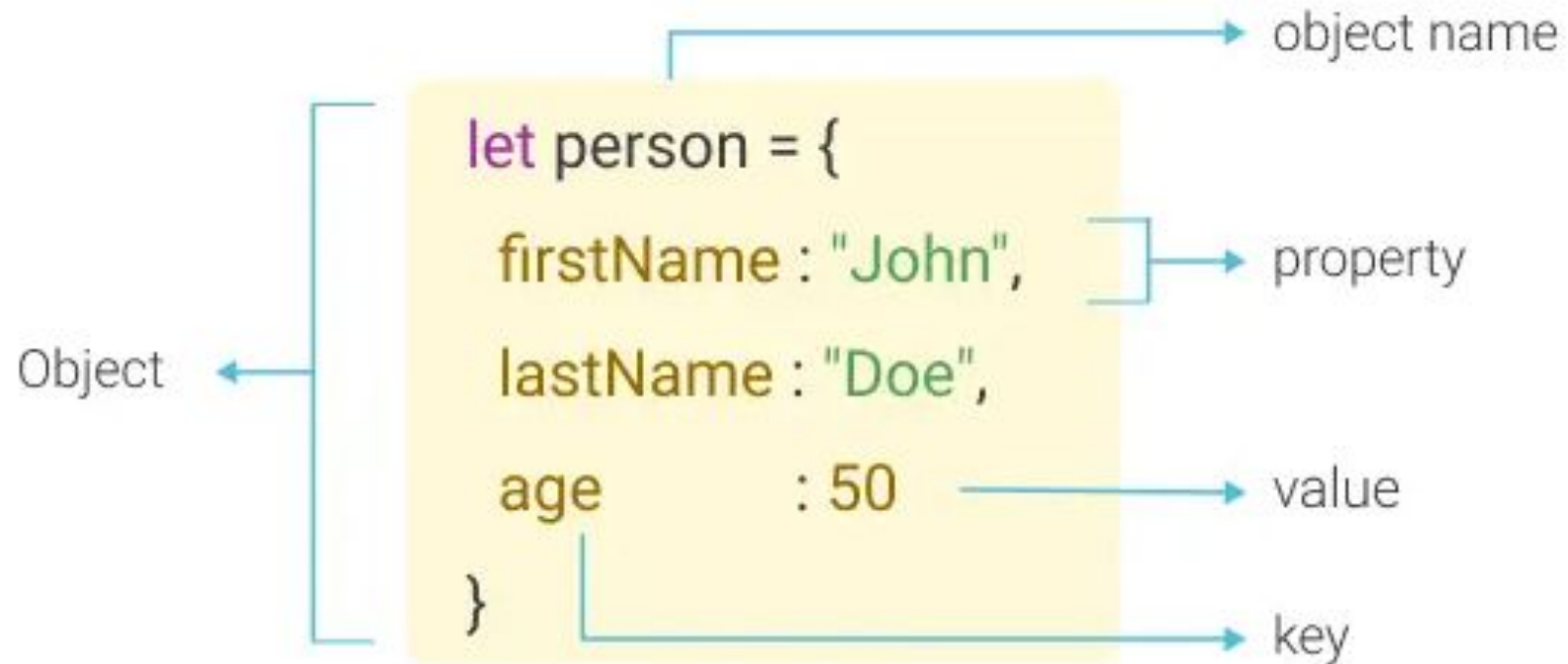
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Objects Oriented Programming

- **Object**
 - **Collection of related entities**
 - **Eg: Ball - Color, Size, Diameter, Texture**
- **OOP**
 - **Way of programming utilizing objects**
- **Objects – Different from C++ and Java**
 - **Does not require a class.**
- **Objects in Javascript**
 - **Array, Object, Math, Date, Regular Expression, and Functions.**

Objects Structure

JavaScript Object Structure



Objects Structure

- **Objects Syntax**

- **identifier_name = { key1 : value1/method1(arguments){method body}, ..
keyn : valuen/methodn(arguments){method body}};**

Creating Objects

- **Objects Syntax**

- **identifier_name = { key1 : value1/method1(arguments){method body}, ..
keyn : valuen/methodn(arguments){method body}};**

- **Eg1:**

```
var student = {  
    "name": "Jane Doe",  
    "class": 2015  
};
```

Creating Objects using dot

- **Eg1:**

```
var student = {  
    "name": "Jane Doe",  
    "class": 2015  
};
```

- **Eg2:**

```
student.name = "Jane Doe"  
student.class = 2015
```

Creating Objects using new

- **Eg1:**

```
var student = {  
    "name": "Jane Doe",  
    "class": 2015  
};
```

- **Eg2:**

```
student.name = "Jane Doe";  
student.class = 2015;
```

- **Eg3:**

```
const student = new Object();  
student.name = "Jane Doe";  
student.class = 2015;
```

Creating Objects using new

- **Eg3:**
const student = new Object();
student.name = “Jane Doe”;
student.class = 2015;
- **new – create String, number, and Boolean objects**

Access Object Members

- **Eg:**

```
var student = {  
    "name": "Jane Doe",  
    "class": 2015  
};
```

- **Access members using dot (.)**

- **student.name;**
- **class.name;**

- **Access members using square brack []**

- **student["name"]**

Access Object Members

- **Eg:**

```
var student = {  
    "name": "Jane Doe",  
    "class": 2015  
};
```

- **Access members using dot (.)**

- **student.name;**
- **class.name;**

- **Access members using square brack []**

- **student["name"]**

- **Accessing non-existent keys is not an error**

- **Cascaded/Nesting objects are possible**

Access Object Members

- **Eg:**

```
var student = {  
    "name": "Jane Doe",  
    "class": 2015  
};
```

- **Preferably - const keyword for objects**

Deleting Objects using new

- Use delete keyword.

- Eg1:

```
var student = {  
    "name": "Jane Doe",  
    "class": 2015  
};
```

- delete student.class;
- delete student[class];

Difference – var, let, const

	Scope	Redeclare	Reassign	Hoisted	Binds this
var	No	Yes	Yes	Yes	Yes
let	Yes	No	Yes	No	No
const	Yes	No	No	No	No

Object Methods

- **Eg:**

```
var student = {  
  "name": "Jane Doe",  
  "greet": function() { console.log('hello')}  
};
```

Object Methods

- **Eg:**

```
var student = {  
  "name": "Jane Doe",  
  "greet": function() { console.log('hello')}  
};
```

- **student.greet()**
- **student.greet – Return function definition**

this

- **this** – refer to data members - inside method
- **Eg:**

```
const person = {  
  name: 'John',  
  age: 30,  
  greet: function() { console.log('The name is' + ' ' + this.name); }  
};
```

Iterate through Object

- **Eg:**
- **const person = {
 name: "John",
 age: 30,
 city: "New York"};**
let txt = "";
for (let x in person) {
txt += person[x] + " ";
};
console.log(txt);
- **in** – iterates through object properties keys.
- **of** - iterates through Arrays, Strings, and so on.

output: John 30 New York

Get Object Values

- **Syntax:**
 - **Object.values(object_literal);**
- **Eg:**
- **const person = {
 name: "John",
 age: 30,
 city: "New York"
};
Object.values(person)**

Best Practices

- 1. Use HTML tags with the same case (preferably lowercase).**
- 2. Filename to be based on functionality.**
- 3. When opening a tag, have a "tab space" for each line until you close the tag. It helps in the readability of your HTML document.**
- 4. Give a comment describing a web page as the first line of your HTML document.**
- 5. Keep HTML, CSS, and Javascript files separate.**
- 6. Give meaningful names to identifiers.**
- 7. Identifiers should not be too long or too short (min 3 character length).**
- 8. Add comments on the outermost nested control structure.**

1. **Paul J Deitel, Harvey M Deitel, and Abbey Deitel, "Internet and the world wide web: How to program," Pearson, Fifth Edition, 2012.**

THANK YOU

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